

CDS DS 121
Foundations of Data Science II
Fall 2026

Lecture Meeting Place and Time: 685-725 Comm Ave CAS B12, MWF 3:35 PM – 4:25 PM

Discussion Meeting Place and Time:

A2: 871 Commonwealth Ave CGS 323,
W 2:30 PM – 3:20 PM

A4: 871 Commonwealth Ave CGS 323,
W 4:40 PM – 5:30 PM

A5: 665 Commonwealth Ave CDS 164,
W 12:20 PM – 1:10 PM

A6: 685-725 Commonwealth Ave CAS 204A,
W 1:25 PM – 2:15 PM

Instructor: Prof. Tanim Chatterjee (tchatt2@bu.edu)

OH: F 12:30PM - 1:30PM (CCDS 1402)

TF: Gavin Rees (gavintr@bu.edu)

OH: M 10AM – 12:00PM

CA: Riwarz Shrestha (riwarz@bu.edu)

OH: W, TR 10:00AM – 11:00AM

Course Description

DS 121 is the second in the three-course sequence (DS 120, 121, 122) that introduces students to theoretical foundations of Data Science. DS 121 covers an introduction to key concepts from Linear Algebra (vector space, independence, orthogonality and matrix factorizations). The DS theme running through the course is exploratory data analysis, enabling a better understanding of the data at hand. The course will link mathematical concepts with computational thinking, specifically through the use of problem sets that require students to answer mathematically-posed questions using computation. This course fulfills a single unit in each of the following BU Hub areas: Quantitative Reasoning I, Digital/Multimedia Expression, Critical Thinking.

Prerequisites

DS 120 (or equivalent) is a prerequisite, and DS 110 (or equivalent) is a corequisite. In other words, you must either have taken DS 110 or be currently registered in DS 110. This course will contain program- ming examples in class and programming assignments in your quizzes. If you have questions about course equivalents, please contact the instructor as soon as possible.

Getting Set Up

You will need access to the following:

Piazza for announcements. Check it weekly!

Link: <https://piazza.com/bu/fall2026/ds121>

If you are not enrolled, use the link to join.

Gradescope for accessing graded exams, assignments and discussion worksheets.

Link: <https://www.gradescope.com>

Entry code: VE8JVK

Blackboard for course materials and resources.

Link: <https://learn.bu.edu/ultra/course>

Python installed on your laptop for practicing coding questions during discussions.

Contact a TF or CA if you need help.

Office Hours

Students are welcome and encouraged to visit office hours. Any changes to the scheduled OH will be announced on Piazza.

Course and Grading Administration

Final grades will be computed based on the following:

25% Midterm1

25% Midterm2

30% Final exam

5% Discussion Pop Quizzes

5% Class Participation

5% Discussion attendance

5% Assignment completion

Overall Course Grade

The instructors will determine curves and grade cutoffs after all assignments, exams and participation scores have been evaluated.

Grade cutoffs will take into account our assessment of the difficulty level of the assignments and exams, and our assessment of what is expected for each letter grade along with overall class performance. The grade boundaries will **not** be disclosed.

*** While the graded midterms will be available to the students for review on Gradescope after grading, the final exams will not be available to the students for review. The students can, however, email the instructor within 48 hours after the final exam scores are released if they feel there is a grading error. ***

Assignments, Quizzes and Participation

There will be weekly assignments released on Gradescope on Fridays (unless otherwise announced in class). Students will have the weekend to look through the questions and think of an approach. There will be classroom discussions on select questions on the assignment the following Monday. Students will be graded for class participation based on these lecture discussions and other questions discussed throughout the lectures (recorded using participation cards.... More to be discussed in class). Students will be required to submit their completed assignments by Wednesday of the following week. They will be graded on completion of assignments and not correctness.

All the solutions to the questions will be posted after the submission deadline on blackboard.

There will also be **5 surprise Pop Quizzes** spread out during discussions. These will be 10 - 15 minute written quizzes covering everything taught in lectures and discussions till date. While attendance in lectures is not compulsory, it is highly encouraged. Students will be graded on class participation. **Attendance** in **discussions** will be recorded and graded. Students will be excused for **at most two** discussion absences without penalty.

Exams:

There will be 3 exams in this course:

Midterm 1: **October 13th, 2026 (Tuesday but a Monday schedule of classes)**

Midterm 2: **November 16th, 2026 (Monday)**

Final: **Week of December 14th – 18th, 2026**

While Midterm 1 and Midterm 2 won't be cumulative, Final will cover everything taught in class throughout the semester.

Note: Please mark these dates in your calendar and make travel plans accordingly.

Collaboration Policy

The goal of the exams is for you to show the instructors what you have learned, so any form of collaboration is strictly prohibited. Computers, phones, and notes are also forbidden during exams unless explicitly stated otherwise. That said, we encourage you to collaborate with classmates when studying lecture materials and preparing for the exams.

Academic Code of Conduct

You must read and adhere to BU's Academic Code of Conduct, which is available here:

<https://www.bu.edu/academics/policies/academic-conduct-code/>. Please familiarize yourself with this code, its definitions of misconduct and plagiarism, and its sanctions.

Violations of this code will result in receiving a score of 0 on the quiz or exam, and may be grounds for referral to BU's Academic Conduct Committee. If you have any questions about the policy, you must ask the instructors in person or via email before taking an action that might be a violation.

Reasonable Accommodations

If you are a student with a disability or believe you might have a disability that requires accommodations, please contact the Office for Disability Services (ODS) at (617) 353-3658 or access@bu.edu to coordinate any reasonable accommodation requests. ODS is located at 25 Buick Street on the 3rd floor.

Learning Environment

This course is committed to fostering an inclusive environment for individuals of all genders, races, cultures, abilities, and sexual orientations. Please treat your instructors and classmates with respect and help create a positive and supportive learning experience for everyone.

Learning Outcomes

This course satisfies 3 Hub requirements: Quantitative Reasoning I, Digital/Multimedia Expression, and Critical Thinking. Each Hub requirement has associated learning outcomes that are listed below.

Quantitative Reasoning I

1. Students will demonstrate their understanding of core conceptual and theoretical tools used in quantitative reasoning, such as statistics, computing, and mathematics.
2. Students will interpret quantitative models and understand a variety of methods of communicating them, such as graphs, tables, formulae, and schematics.
3. Students will communicate quantitative information symbolically, visually, numerically, or verbally.
4. Students will recognize and articulate the capacity and limitations of quantitative methods and the risks of using them improperly.

This class will introduce students to core tools used in quantitative reasoning, including vector spaces, independence, matrix decomposition, dimensionality reduction and linear systems. The quizzes will give students the chance to use computational tools to manipulate and make sense of data, communicate conclusions based on mathematical and computational representations, and identify misrepresentations based on incorrect use of the tools.

Digital/Multimedia Expression

1. Students will be able to craft and deliver responsible, considered, and well-structured arguments using media and modes of expression appropriate to the situation.

2. Students will be able to demonstrate an understanding of the capabilities of various communication technologies and be able to use these technologies ethically and effectively.
3. Students will be able to demonstrate an understanding of the fundamentals of visual communication, such as principles governing design, time-based and interactive media, and the audio-visual representation of qualitative and quantitative data.

Through the lectures and in quizzes, we will explore how to visually represent data in ways that support logical reasoning. Students will take complex multidimensional data and communicate arguments and interpretations using strategies such as projections, compression, clustering, and summarization.

Critical Thinking

1. Students will be able to identify key elements of critical thinking, such as habits of distinguishing deductive from inductive modes of inference, recognizing common logical fallacies and cognitive biases, translating ordinary language into formal argument, distinguishing empirical claims about matters of fact from normative or evaluative judgments, and recognizing the ways in which emotional responses can affect reasoning processes.
2. Drawing on skills developed in class, students will be able to evaluate the validity of arguments, including their own.

Students will learn to think deeply about their data, to understand its properties, and based on the initial analyses translate their intuitive understanding of their data into formal problem definitions. Students will be able to evaluate whether models provide a good fit to data, and whether a given model is valid as a description of data.

Topics to be covered

Tentative...Not written in stone!!!

- Introduction
- Review of vectors

- Lengths and Inner products
- Intro to KNN
- Model Selection
- K-means and its workflow
- Analysis of K-means
- Linear Systems review
- Linear dependence
- Matrices and linear systems
- Computational Complexity
- Transformations
- Visualizing Transformations
- Inverses and Determinants
- Elementary Matrices
- LU Factorization
- Column and Null Spaces
- Basis and dimensions
- Co-ordinate systems
- Orthogonal projections and Gram-Schmidt Orthogonalization
- QR Factorization
- Least Squares
- Linear Regression
- Markov Chains
- Eigenvalues and Eigenvectors
- Properties of eigenvalues and eigenvectors
- Diagonalization
- SVD